

Note: The following is a Beamer presentation from a keynote address I gave at Kathmandu University in 2025. Slide 21 contains an open problem, and slide 25 contains two open problems. To understand the context you should probably go through the presentation from the beginning.

Zeros of harmonic polynomials

Michael Dorff

President of the Mathematical Association of America, 2019-2021

Executive Director of TPSE Math, 2021-2024

Professor of Mathematics

Brigham Young University

Utah, USA



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This is a real polynomial. **How many zeros does it have?**

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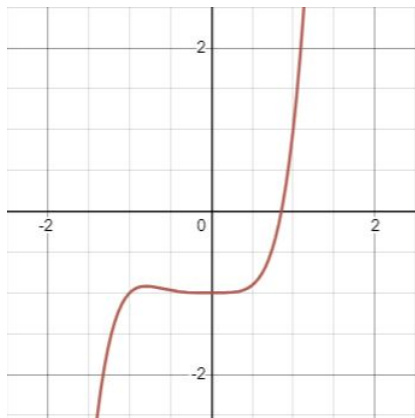
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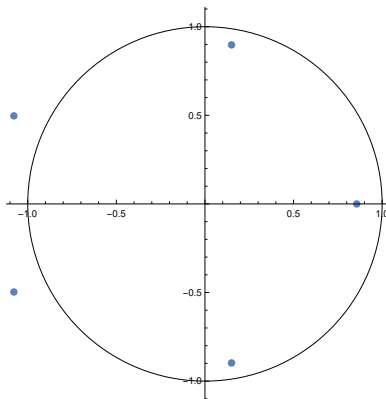
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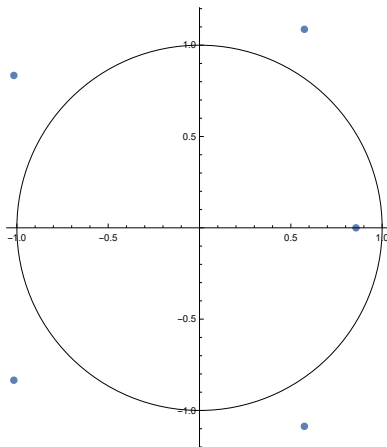
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Let's increase the coefficient of $\bar{z}^4$. **Does it have 5 zeros?**

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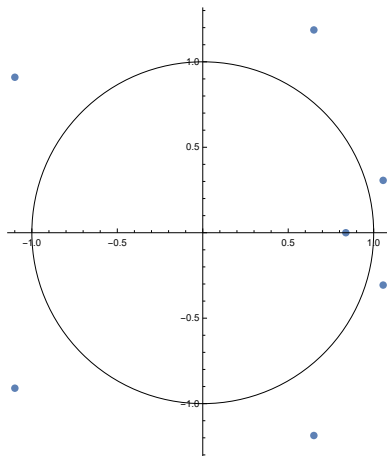
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It has 7 zeros!

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What happens to the number of zeros?

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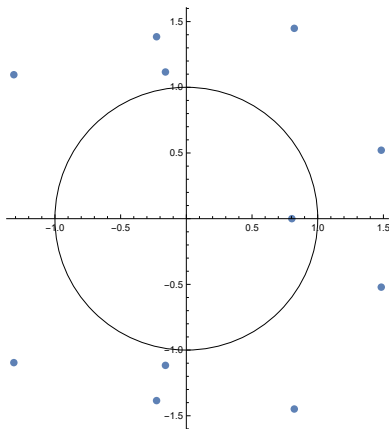
We now have 11 zeros!

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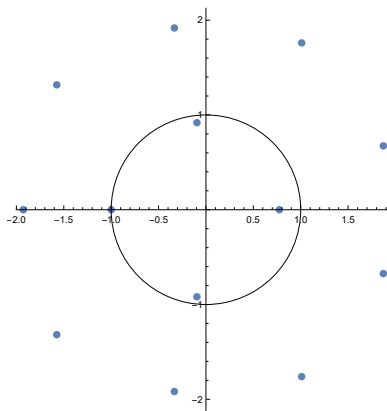
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The FTA does not describe complex-valued harmonic polynomials.

Problem

What are the maximum number of zeros of a harmonic polynomial?

$$p(z) = a_n z^n + \cdots + a_0 + b_n \bar{z}^k + \cdots + b_1 \bar{z}$$

Definition (Analytic)

Let $F : D \subset \mathbb{C} \rightarrow \mathbb{C}$. The function $F(x, y) = u(x, y) + iv(x, y)$ is an analytic function if:

- F is continuous;
- u and v are real harmonic in D ; and
- u and v are harmonic conjugates (that is, $u_x = v_y$ and $u_y = -v_x$).

Harmonic Functions

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Definition (Harmonic)

Let $f : D \subset \mathbb{C} \rightarrow \mathbb{C}$. The function $f(x, y) = u(x, y) + iv(x, y)$ is a (complex-valued) harmonic function if:

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Theorem (Clunie, Sheil-Small)

If $f = u + iv$ is harmonic in a simply-connected domain G , then

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Problem

Let $p = h + \bar{g}$ be a harmonic polynomial where the degree of h is n , the degree of g is k , and $n \geq k$:

$$p(z) = a_n z^n + \cdots + a_0 + b_k \bar{z}^k + \cdots + b_1 \bar{z}$$

What are the maximum number of zeros of p ?

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Conjecture (Sheil-Small)

Maximum number of zeros of p is n^2 .

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However, Lee et. al. showed Wilmhurst's conjecture is not true in general.

Our Work: - Brilleslyper, Brooks, Dorff, Howell, Schaubroeck

We investigated the maximum number of zeros
for the family of harmonic trinomials

$$p_c(z) = z^n + c\bar{z}^k - 1,$$

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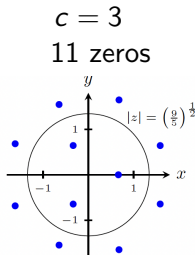
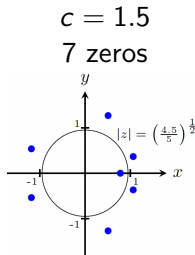
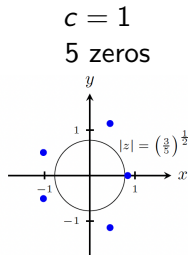
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Let $p_c(z) = z^n + c\bar{z}^k - 1$ where $n - 1 \geq k \geq 1$ and let $N = \lfloor \frac{k}{2} \rfloor + 1$.
There exist N critical values c_j , with $0 < c_1 < c_2 < \dots < c_N$, such that

- (i) if $0 < c < c_1$, then $p_c(z)$ has n distinct zeros,
- (ii) if $c_j < c < c_{j+1}$ for some $1 \leq j \leq N - 1$, then $p_c(z)$ has $n + 4j - 2$ distinct zeros, and
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- (iii) $c > c_2$, $p_c(z)$ has $5 + 2(3) =$ **11 zeros**

Complex Dilatation and Regions of Zeros

Definition: The *complex dilatation* $\omega(z)$ of $f(z) = h(z) + \overline{g(z)}$ is

$$\omega(z) = \frac{g'(z)}{h'(z)}$$

- The sense-preserving region is the region where $|\omega(z)| < 1$.
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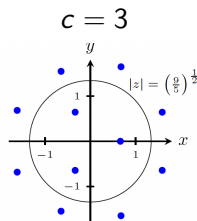
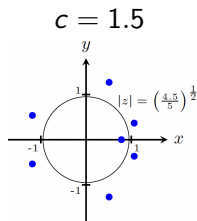
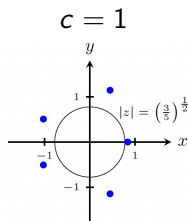
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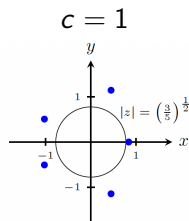
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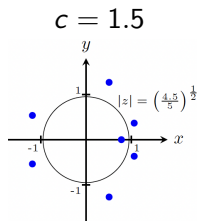
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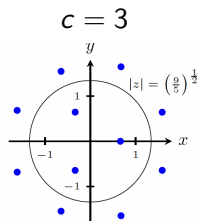
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7 total zeros

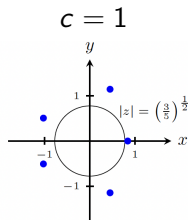


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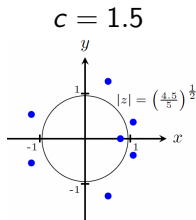
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5 total zeros

5 pos zeros

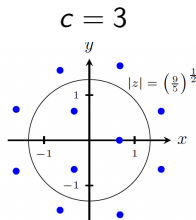
0 neg zeros



7 total zeros

6 pos zeros

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11 total zeros

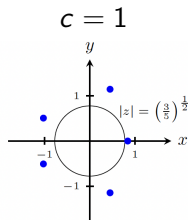
8 pos zeros

3 neg zeros

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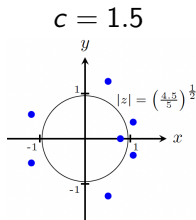
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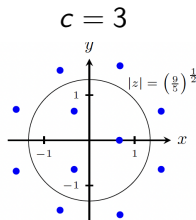
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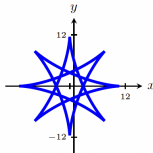
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Definition: A hypocycloid centered at the origin is the curve traced by a fixed point on a circle of radius b rolling inside a larger origin-centered circle of radius a . The curve is given by the parametric equations

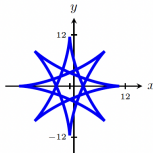
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(8,5) hypocycloid

Lemma 3: The image of the critical circle, $p_c(\Gamma_c)$, is an $(n + k, n)$ hypocycloid centered at $z = -1$. The value of c affects only the size of the hypocycloid.

Lemma 4: The hypocycloid, $p_c(\Gamma_c)$, has $N = \lfloor k/2 \rfloor + 1$ distinct intersections with the real axis to the right of its center. These intersections, in turn, correspond to the N critical c -values, c_j with $0 < c_1 < \cdots < c_N$, at which $p_c(\Gamma_c)$ intersects the origin. The winding number, W_c , of $p_c(\Gamma_c)$ around the origin is as follows:

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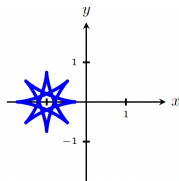
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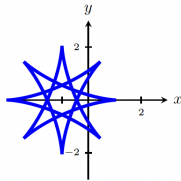
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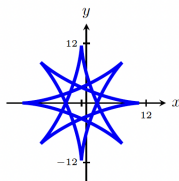
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Summary: Theorem [BBDHS]

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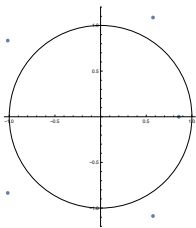
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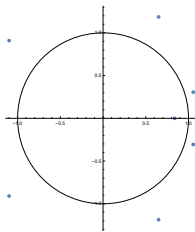
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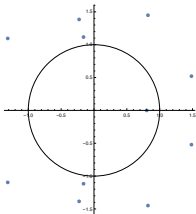
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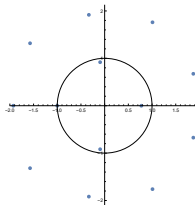
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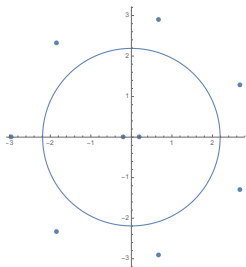
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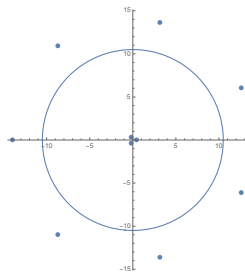
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$$p_{c_1}(z) = z^5 + \frac{105}{4} \bar{z}^2 - 1$$

9 roots



$$p_{c_2}(z) = z^4 + 14 \bar{z}^3 - 1$$

10 roots

Question: For $s \in [0, 1]$, what is the max number of zeros of

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Could it be 10? **Yes, but . . .**

Could it be larger than 10? **Yes. In fact, it is 11!**

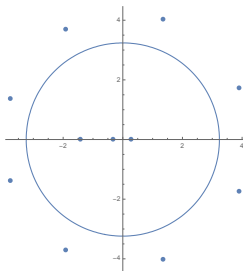
Question: For $s \in [0, 1]$, what is the max number of zeros of

$$\begin{aligned} P(z) &= s [p_{c_1}(z)] + (1 - s) [p_{c_2}(z)] \\ &= s [z^5 + c_2 \bar{z}^2 - 1] + (1 - s) [z^4 + c_1 \bar{z}^3 - 1]? \end{aligned}$$

Could it be 9? **No**

Could it be 10? **Yes, but . . .**

Could it be larger than 10? **Yes. In fact, it is 11!**



$$P(z) = \frac{4}{9} \left[z^5 + \frac{105}{4} \bar{z}^2 - 1 \right] + \frac{5}{9} \left[z^4 + 14 \bar{z}^3 - 1 \right]$$

Theorem

Let

$$P(z) = s(z^m + c_1 \bar{z}^k - 1) + (1 - s)(z^n + c_2 \bar{z}^\ell - 1).$$

Let $m, k, n, \ell \in \mathbb{N}$ satisfy $m + k = n + \ell$ and $m > n > \ell > k$.

Let $s = \frac{n}{m+n}$ and $c_1 = \frac{mbl}{kn}$.

Then, there exists c such that for all $c_2 > c$, P has $m + 2\ell$ zeros.

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Open Problems:

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- 2 Does anything interesting happen when we consider a convex combination of three terms such as

$$\begin{aligned} f(z) = & s_1 (z^{m_1} + a_1 \bar{z}^{k_1} - 1) \\ & + s_2 (z^{m_2} + a_2 \bar{z}^{k_2} - 1) \\ & + s_3 (z^{m_3} + a_3 \bar{z}^{k_3} - 1), \end{aligned}$$

where $0 \leq s_j \leq 1$ for $j = 1, 2, 3$ and $s_1 + s_2 + s_3 = 1$?



Utah in USA



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Utah has Beautiful Outdoors

Thank you!

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